BIPHMUN 2026

Background Guide
Committee: United Nations Environment Programme
Topic: Addressing the Environmental and Health Impacts of AI and Crypto Related E-Waste
Agenda Overseen by: Mia Chen

1. Description of the Issue

1.1 History of the Issue

The whole conversation about Artificial Intelligence (AI) can be traced back to early work on computer science and mathematics. Boolean algebra, also known as binary algebra, is a mathematical system that operates on binary variables, where each variable can be either true (1) or false (0) (Snyder, 2024). Boolean algebra laid the early foundation for computers and computer science.

Long before the term Artificial Intelligence was coined, a man called Alan Turing proposed the well-known Turing Test in 1950, used for determining if a machine is capable of thinking (Turing, 1950).

The concept and the field of AI is officially born in the Dartmouth Summer Research Project on Artificial Intelligence in 1956. It was a two-month and ten-man project with the goal of training machines to learn and use language; complex concepts and abstract thinking; solve problems and improve themselves (J. McCarthy et al., 1955). Although the project did not produce a major output, it laid the foundation of fields such as machine learning and language processing.

In the decades that followed, AI research advanced through several periods of growth and decline. Early programs such as ELIZA, developed by Joseph Weizenbaum in 1966, was one of the first computer programs designed to process natural language; it demonstrated the potential for computers to simulate human conversation and laid foundation for more advanced projects such as the Large Language Models (LLMs) we are currently using (London Intercultural Academy, 2024).

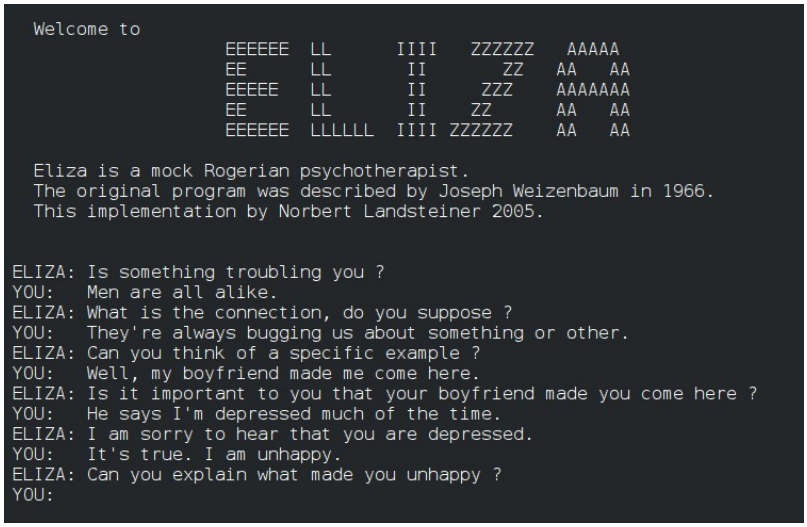

Figure 1 The User Interface of ELIZA

After ELIZA, the field entered a period of "AI Boom," shortly after when The American Association of Artificial Intelligence was formed. The time period witnessed the development of AARON, an autonomous drawing program and the commercial launch of Alacrity, the first strategy managerial advisory system aided by a complex expert system, by Alactrious Inc. However, an AI winter soon followed the boom. The AI winter was a period characterized by low consumer, public, and private interest in AI, leading to less fundings and breakthroughs (tableau).

Despite the lack of funding, the early 90s witnessed huge improvements in the field of AI, starting with the legend of IBM's Deep Blue. In 1997, IBM's Deep Blue, a chess machine, defeated the world chess champion Gary Kasparov (Deep Blue | IBM, 2025).

Figure 2 The Match between Kasparov and Deep Blue

Parallel to the advancement of AI and drawdowns in the financial system, the concept of cryptocurrency was introduced to the public. In 2009, the era of cryptocurrency formally began with the commercial launch of Bitcoin after Satoshi Nakamoto's publishing of the Bitcoin Whitepaper. Bitcoin uses a Proof of Work (PoW) system, which required computers to use brute-force algorithms to verify if a transaction data, appended with a "nonce", would produce a hash output starting with 30 or so leading zeros (Nakamoto, 2009). Application-Specific Integrated Curcuits (ASICs) were developed to specialize in this task; however, ASICs' inflexibility in repurposing means older versions would be obsolete once newer versions come out (CryptoDamus, n.d.)

On the other hand, the world saw more advancements in application of AI. In 2011, Apple launched Siri as a virtual assistant. In 2016, Hanson Robotics created Sophia, the "robot citizen" with face and limbs resembling those of humans. In November 2022, the phenomenal ChatGPT-3, a model using Deep Learning to finish assigned tasks, was launched. Now, we are experiencing an "LLMs Boom," where LLMs such as ChatGPT, Gemini, and Deepseek aid our daily tasks (Stryker, 2023).

Modern artificial intelligence systems, particularly Large Language Models (LLMs), are typically trained using deep learning techniques. These models learn by analyzing extremely large datasets and repeatedly adjusting millions or even billions of internal parameters in order to improve prediction accuracy. During training, the model processes massive amounts of text or data many times through mathematical operations such as matrix multiplications. Because these computations can be

performed simultaneously across many data points, specialized hardware is required to handle the workload efficiently. As a result, developers rely heavily on Graphics Processing Units (GPUs), which are designed to perform thousands of calculations in parallel. Compared with traditional central processing units (CPUs), GPUs can process the large-scale numerical operations required for deep learning far more quickly, making them essential for training modern AI systems.

Figure 3 Examples of LLMs

In the mean time, the advancement of AI and cryptocurrencies is accompanied by increasing waste production. In Bitcoin mining, one tends to use cutting-edge hardware to be competitive, leading to rapid hardware evolution and improper recycling. Bitcoin mining has been found to consume as much energy as small countries, which translates into a significant carbon footprint (de Vries & Stoll, 2021). As of May 2021, Bitcoin's annual waste production reached 30.7 metric kilotons.

A similar pattern can be observed in the development of artificial intelligence systems. Training modern AI models requires large amounts of specialized computing hardware, particularly Graphics Processing Units (GPUs). Technology companies and research institutions often upgrade their hardware frequently in order to train larger and more complex models. As newer and more powerful processors are released, older hardware quickly becomes obsolete or less economically viable to operate. This rapid cycle of technological advancement contributes to the growing accumulation of electronic waste associated with AI infrastructure. Researchers predict that AI-related E-waste could reach "a total accumulation of 1.2–5.0 million tons during 2020–2030" (Wang, P., Zhang, LY., Tzachor, A. et al.).

Electronic waste also poses serious health risks, particularly in regions where informal recycling practices are common. Workers often dismantle electronics manually or

burn components to recover valuable metals such as copper and gold. These processes release toxic substances including lead, mercury, and cadmium into the air, soil, and water. According to the World Health Organization, exposure to these substances can cause neurological damage, respiratory illnesses, developmental disorders in children, and other long-term health problems. As the demand for AI hardware and cryptocurrency mining equipment grows, the volume of discarded electronics entering these recycling systems may also increase, raising concerns about public health impacts in affected communities.

The issue, however, led to more complicated debates beyond environmental and health effects. As policies are recommended to tackle the environmental effects, it is inevitable that they will pose changes on the supply-side of the industries. Although the UNEP is not a binding committee like the Security Council, the policy outlines in its passed resolutions may eventually trickle down to detailed implementations either by other UN agencies or by politicians around the world. This may pose different effects on other countries. For example, AI systems rely on expensive computing infrastructure such as data centers and GPUs. When regulations restrict hardware supply or increase costs, the impact spreads across the entire technology sector. Increased chip prices will raise the cost of data centers, which will influence the cost of software companies, leading to increased service prices. The effects of supply-side regulation can also be found in the use of export control as political leverage. Recent policy debates in the U.S. consider limiting U.S. exports of advanced AI-related hardware produced by NVIDIA, influencing technological competition globally.

However, the costs of not solving the issue is also paramount, including chronic environmental and health damage and widening gaps between the Global North, producing these E-Wastes, and the Global South, receiving these E-Wastes.

1.2 Definition of Key Terms

Aritificial Intelligence (AI): a computational system trained and used to perform human-level tasks.

Electronic Waste (E-Waste): waste typically associated with discarded electronic devices.

Large Language Models (LLMs): lanaguge models trained with self-supervised machine learning on a vast (and usually selected) amount of text.

Cryptocurrency: a digital currency exists only electronically, with a major one being Bitcoin.

Carbon Footprint: the total amount greenhouse gases generated by human actions.

ASICs (Application-Specific Integrated Circuits): Specialized chips used in Bitcoin mining that cannot be repurposed for anything else once they are obsolete.

GPUs (Graphics Processing Units): Versatile chips used for AI training; while more reusable than ASICs, their high-intensity use leads to faster physical degradation.

1.3 Past International Actions

The Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal was adopted in 1989 and it came into force in 1992. It is a global environment agreement on hazardous wastes and other wastes. The Basel Convention regulates transboundary movements of hazardous wastes and implores its member countries to manage their waste production and disposal in an environmentally sound manner (Environment Programme, 2025).

Later, seeing the impact and damage of the climate change, the Paris Agreement was signed in 2015. The Paris Agreement is a legally binding international treaty on climate change. It was adopted by 195 Parties at the UN Climate Change Conference (COP21) in Paris, France. It entered into force on 4 November 2016. Currently, as of 27 January 2026, there are 194 Parties to the Paris Agreement (UNFCCC, 2025).

Since 2016, the United Nations has been issuing e-waste monitor of countries, regions, and the globe. Despite the surge of e-waste, there is not much proper recycling or processing currently, as on 22% of 2022's e-waste has been properly collected and recycled (UNITAR, 2024). The ILO and WHO estimate that millions of

women and children are involved in informal recycling activities, leading to adverse health impacts (World Health Organization, 2024).

2. Bloc Positions

2.1 Stance of Developed Countries

United States:

As a major technology and finance hub, United States, while having concerns over the environmental and health impacts of e-waste, may put the focus on developing AI and cryptocurrencies. As of 2025, 26 states have enacted statewide e-waste laws, depite not having a federal one.

China:

The AI industry in China is robust, too. China gained its prestige in the said field soon after the successful launch of DeepSeek. As one of the leaders of the industry, China will also put its focus on the growth of the industry. Meanwhile, China is also taking actions to mitigate the environmental impacts of e-waste. In fact, in 2018, China impletemented a waste import ban. China is also the first developing country to establish a waste electrical and electronic equipment (WEEE) management fund policy, made official in 2012 (Deng et al., 2024).

European Union (EU):

The EU was the first body to issue a comprehensive act about AI. In the 2024 EU Artificial Intelligence Act, the EU classifies and makes rule of AI according to its risks, with manipulative AI banned, and other high-risk AI strictly regulated (EU AI Act, 2024). In 2012, the EU published the Waste Electrical and Electronic Equipment (WEEE) Directive, putting the prevention of the creation of WEEE as its priority (European Commission, 2025). It also promotes the reuse and recycling of WEEE.

Kenya:

Kenya is the major e-waste generating country in East Africa as the regional hub of digital services (Second Hand Electronics Boom Deepens Kenya's E-Waste Dilemma | Bamako Convention, 2026). Recognizing the issue, Kenya established the National Environment Management Authority (NEMA), and drafted the E-Waste Management regulations in 2013.

Other Developed Countries:

The developed countries typically already have laws and directives regarding AI and e-waste in force. Therefore, they tend to prioritize sustainable development of the AI and cryptocurrency industry. They also will prioritize the independent development of AI to reduce their dependence on other tech giants like the United States.

Other Developing Countries:

They are the major victims of e-waste pollution. Tech giants usually dump their waste into small, developing countries. They are likely against the expansion of the AI industry, and will put the focus on the environmental and health impacts of the industry.

3. Possible Solutions

Possible solutions can be categorized as upstream, usage, and downstream.

In the upstream, solutions mainly tackle the design and manufacturing stage of AI- and Crypto- related hardware. Possible approaches include longer-lasting chips and inter-changeability of hardware. Countries could also implement Extended-Producer Responsibility (EPR) where companies must manage the disposal of their hardware. They could also be designed to be more recyclable. Many dispose of and recycle them inefficiently. Therefore, delegates may consider forming a collaboration to improve the design of the said hardware, and make the manufacturing and recycling process more sustainable. However, the capital expenditure needed for more efficient and long-lasting hardware maybe high, and progress may eventually plateau.

Regulations could also be imposed in the usage stage of AI and Crypto infrastructure. For example, efficiency standards could be implemented for data centers, and limits can be placed on extremely energy-intensive mining operations for Crypto. However, these may vary between countries; some may be more inclined to loosen standards for higher productivity.

In the disposal stage, formal recycling infrastructure can be effective for mitigating the produced e-wastes. Countries could support recycling programs in developing countries and setting up tracking systems for international flows of electronic waste. However, these may be politically sensitive and implementationally difficult. For the "victim" countries, delegates may consider implementing an importation ban, which may temporarily mitigate the mounting of e-waste. However, delegates should be aware of the cost of this decision.

Delegates may also gain inspiration from the carbon tax, where firms are charged with tax proportional to their carbon emissions. By taxing countries or firms, less e-waste production is expected.

4. Evaluation

In short, delegates should consider both the manufacture and the disposal process of machine learning and cryptocurrency mining. Also, delegates should be aware of the competition among tech giants as they form blocs and write resolutions.

There are not yet many direct past international efforts regarding this problem. Delegates are advised to learn from past indirect international efforts when drafting resolutions.

Delegates' performances are monitored throughout the conference. Those who did extensive prior research, collaborate well with others in blocs, give constructive insights in resolutions, and ask insightful questions to other delegates are likely to be recognized and respected. The beginners will not be overlooked. Detailed guidance and assistance would be provided in necessary circumstances. We would like to witness fruitful personal growth and a harmonious environment. See you guys in BIPHMUN 2026!

5. Bibliography

J. McCarthy, M. L. Minsky, N. Rochester, & C.E. Shannon. (1955, August 31). *A PROPOSAL FOR THE DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL I*. http://jmc.stanford.edu/articles/dartmouth/dartmouth.pdf

Snyder, S. (2024). *Boolean algebra | EBSCO*. EBSCO Information Services, Inc. | Www.ebsco.com. https://www.ebsco.com/research-starters/science/boolean-algebra

Turing, A. (1950). Computing Machinery and Intelligence. *Mind*, *59*(236), 433–460. https://doi.org/10.1093/mind/LIX.236.433

London Intercultural Academy. (2024). *The Story Of ELIZA: The AI That Fooled The World*. London Intercultural Academy. https://liacademy.co.uk/the-story-of-eliza-the-ai-that-fooled-the-world/

What is the history of artificial intelligence (AI)? (n.d.). Tableau. https://www.tableau.com/data-insights/ai/history#ai-maturation

Deep Blue | IBM. (2025). Ibm.com. http://ibm.com/history/deep-blue

Stryker, C. (2023, November 2). *What are large language models (LLMs)?* Ibm.com; IBM. https://www.ibm.com/think/topics/large-language-models

EU AI Act. (2024, February 27). *High-level Summary of the AI Act*. EU Artificial Intelligence Act; Future of Life Institute. https://artificialintelligenceact.eu/high-level-summary/

UNFCCC. (2025). *The Paris agreement*. United Nations Climate Change. https://unfccc.int/process-and-meetings/the-paris-agreement

UNITAR. (2024). *The global E-waste Monitor 2024 – Electronic Waste Rising Five Times Faster than Documented E-waste Recycling: UN*. E-Waste Monitor. https://ewastemonitor.info/the-global-e-waste-monitor-2024/

World Health Organization. (2024, October 1). *Electronic Waste (E-Waste)*. World Health Organization; World Health Organization. https://www.who.int/news-room/fact-sheets/detail/electronic-waste-(e-waste)

European Commission. (2025). *Waste from Electrical and Electronic Equipment (WEEE)*. European Commission. https://environment.ec.europa.eu/topics/waste-and-recycling/waste-electrical-and-electronic-equipment-weee_en

Second hand electronics boom deepens Kenya's e-waste dilemma | Bamako Convention. (2026, January 8). Bamakoconvention.org. https://www.bamakoconvention.org/node/86

de Vries, A., & Stoll, C. (2021). Bitcoin's growing e-waste problem. Resources, Conservation and Recycling, 175, 105901. https://doi.org/https://doi.org/10.1016/j.resconrec.2021.105901

Deng, Y., Wu, W., Zhang, X., Li, S., Song, X., & Wang, J. (2024). Overview of China's Waste Electrical and Electronic Equipment Recycling in the Last Two Decades. Sustainability, 16(23), 10683. https://www.mdpi.com/2071-1050/16/23/10683

Nakamoto, S. (2009). Bitcoin : A peer-to-peer electronic cash system. 61, 200–202. https://bitcoin.org/bitcoin.pdf

The Mining-to-AI Conversion Challenge: Infrastructure, costs & realities | CryptoDamus. (n.d.). Cryptodamus. https://cryptodamus.io/en/articles/news/mining-to-ai-why-the-pivot-is-more-complex-than-you-think

Wang, P., Zhang, LY., Tzachor, A. et al. E-waste challenges of generative artificial intelligence. Nat Comput Sci 4, 818–823 (2024). https://doi.org/10.1038/s43588-024-00712-6